Implementation of Multiplexer and De-Multiplexer

Objectives:

1. To understand the working principle of Multiplexer and De-Multiplexer circuits and how they control the selection and distibution of data using select lines.
2. To design and implement Multiplexer and De-Multiplexer circuits using logic gates and verify their operation through truth tables.

Basic Theory:

1. Multiplexer:

Multiplexer means many into one. A Multiplexer is a circuit used to select and route any one of the several inputs signals to a single output.

It handle two type of data that is analog and digital.

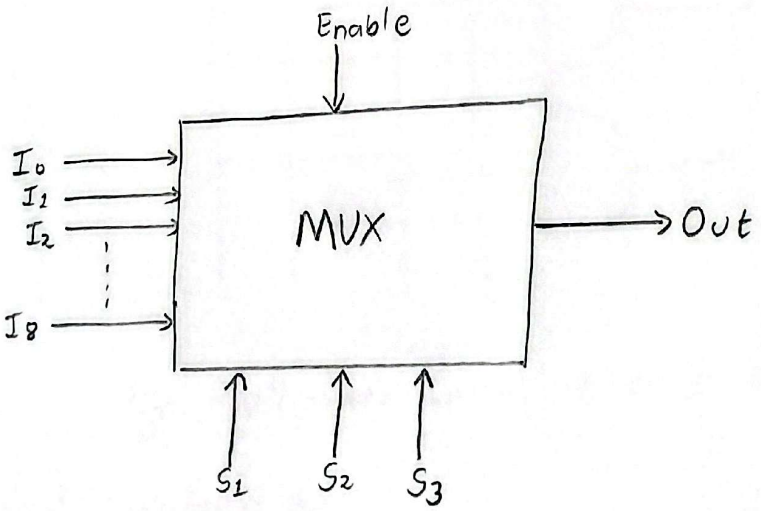


fig. Block Diagram of MUX (8 to 1)

Logic Diagram :

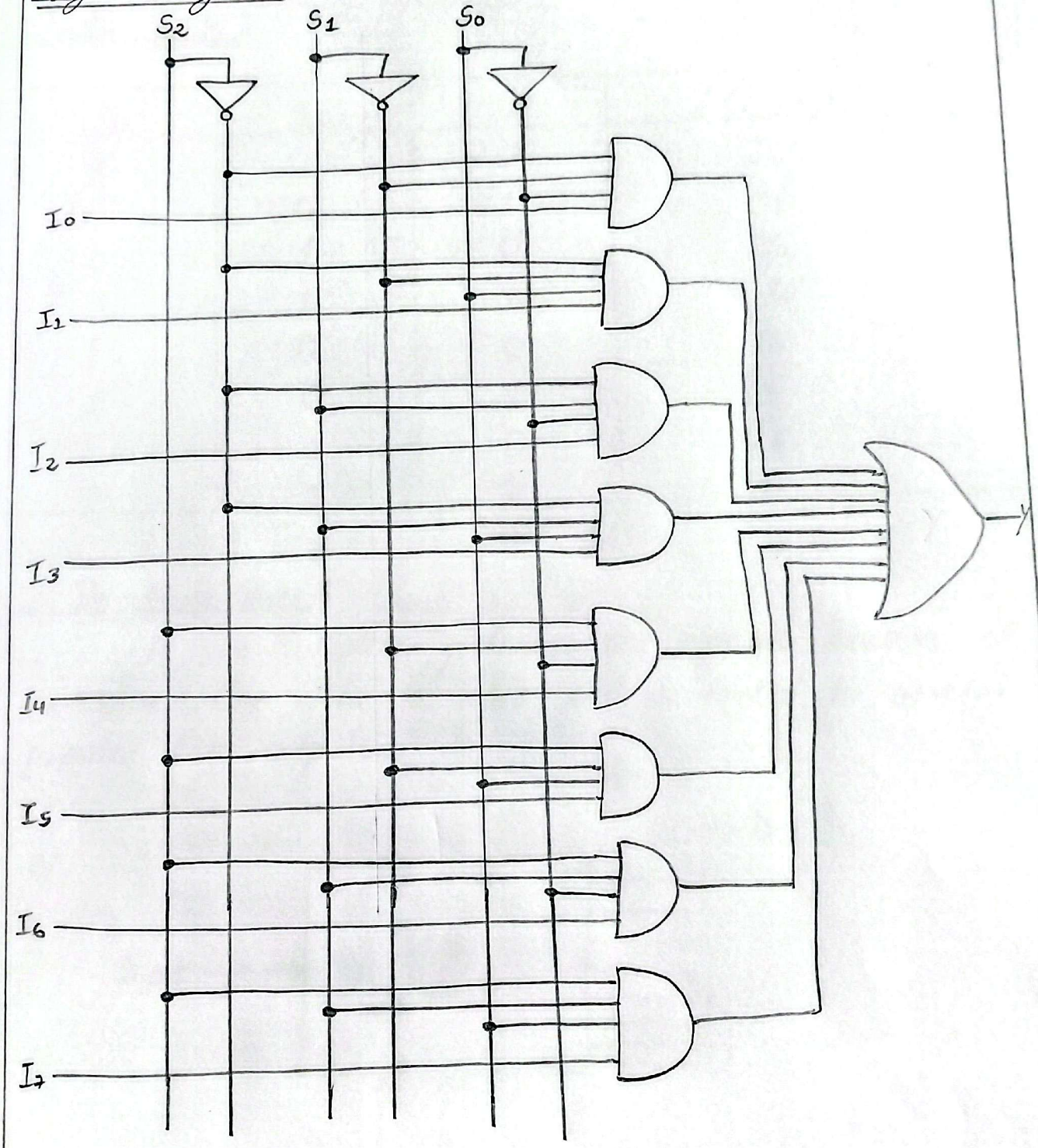


fig. Multiplexer Circuit 8 to 1.

The Output calculated as

$$y = S_2'S_1'S_0'I_0 + S_2'S_1'S_0I_1 + S_2'S_1S_0'I_2 + S_2'S_1S_0I_3 + S_2S_1'S_0'I_4 + S_2S_1'S_0I_5 + S_2S_1S_0'I_6 + S_2S_1S_0I_7$$

Truth Table:

S_2	S_1	S_0	Y (Output)
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

2. De-Multiplexer:

A De-Multiplexer performs the opposite function of multiplexer. One data is input and is routed to several possible data outputs.

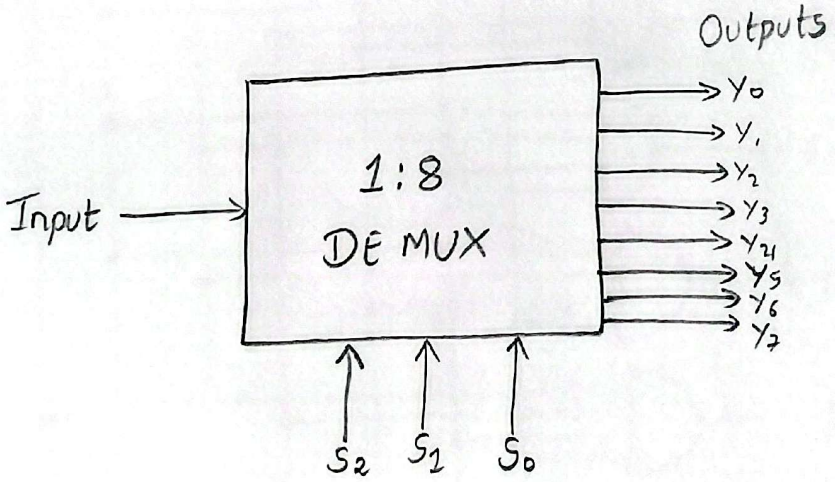
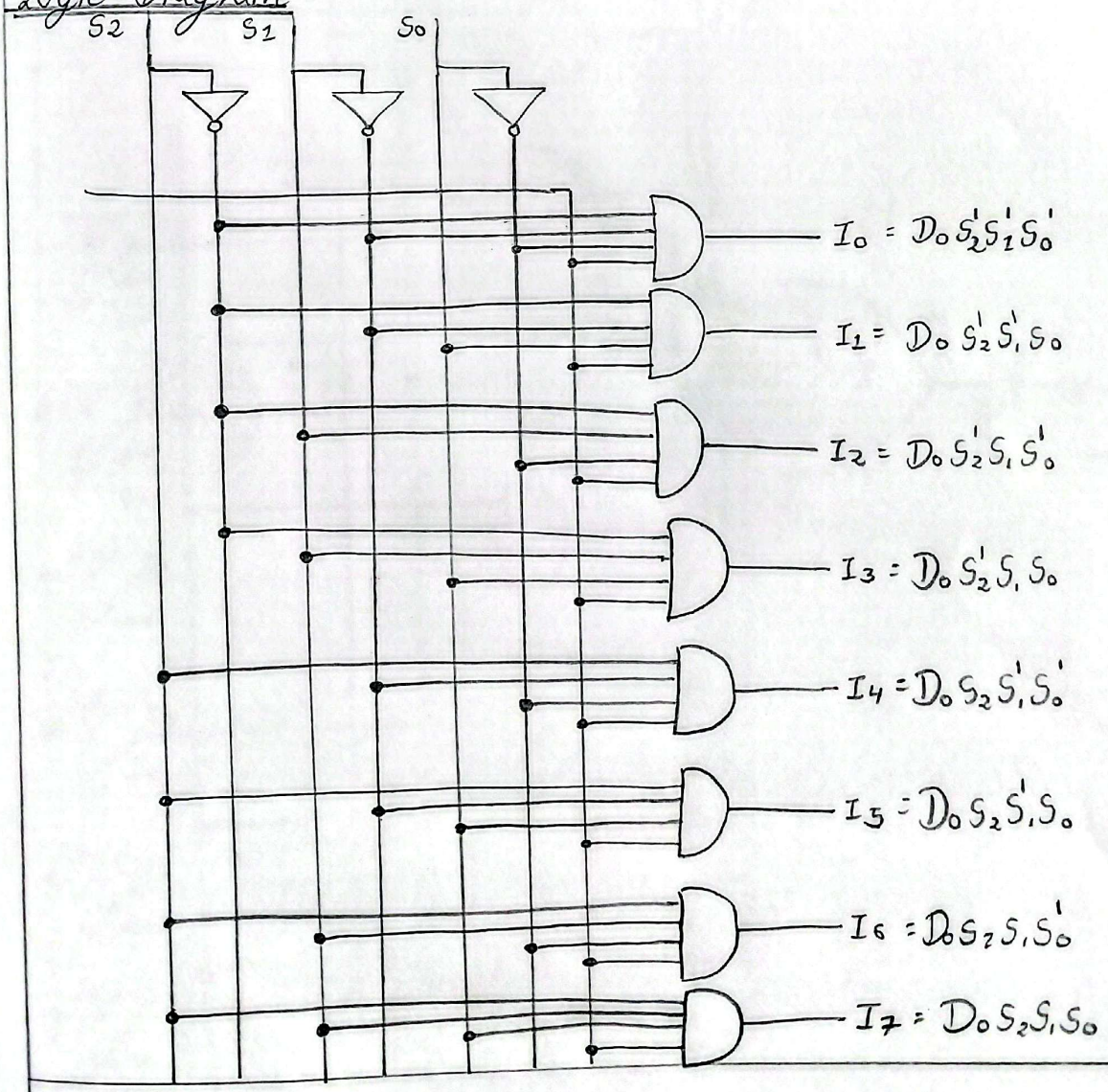


fig. Block Diagram of 1:8 De-Multiplexer

Truth Table:

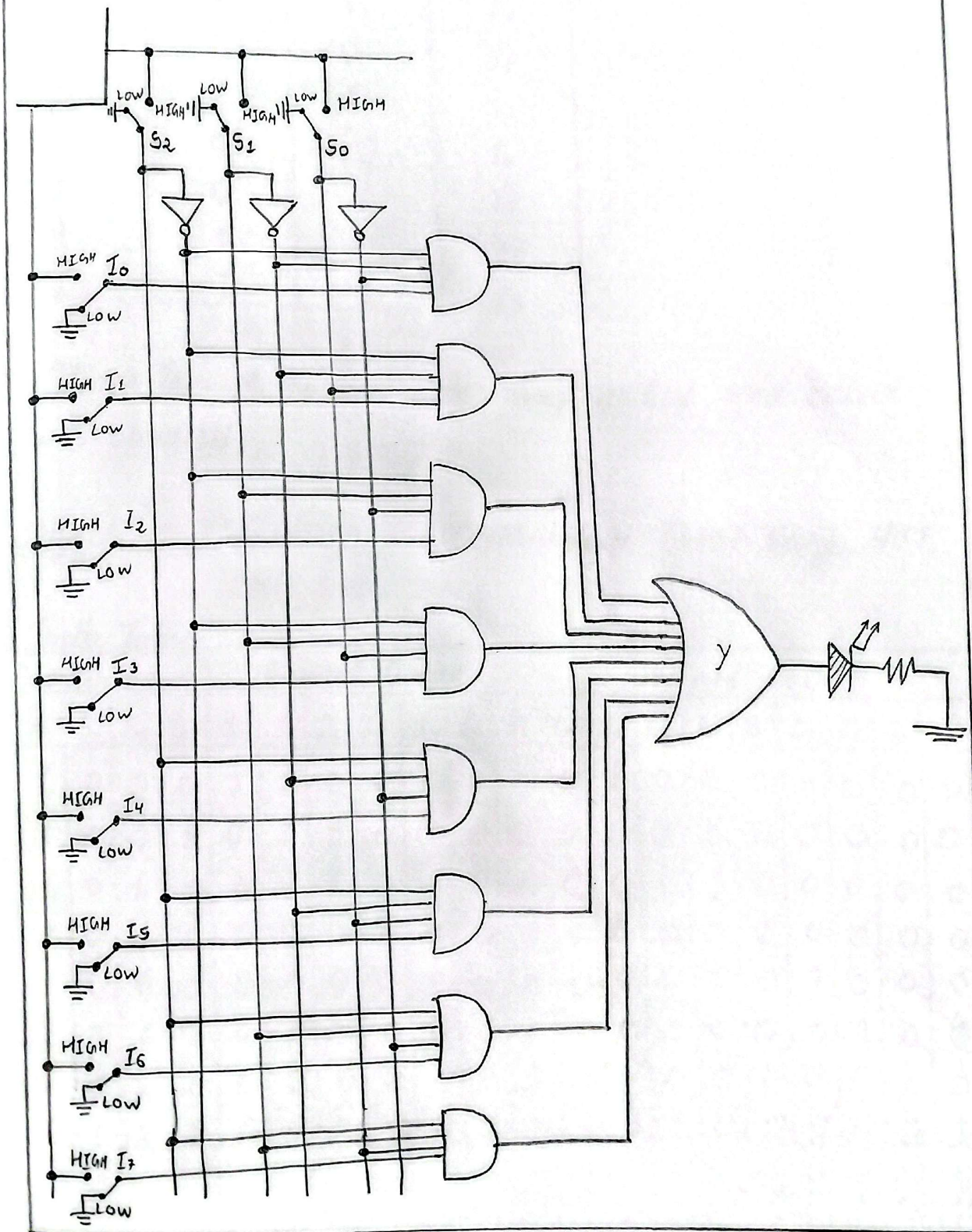
Input D_0	S_2	S_1	S_0	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	1

Logic Diagram:



Diagrams and Observations:

Objective 1: Building a Multiplexer circuit using NOT, AND OR gates.



Truth Table:

INPUTS											OUTPUTS	
S_2	S_1	S_0	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	Expected(Y)	Observed(Y)
0	0	0	0	x	x	x	x	x	x	x	0	0
0	0	0	1	x	x	x	x	x	x	x	1	1
0	0	1	x	0	x	x	x	x	x	x	0	0
0	0	1	x	1	x	x	x	x	x	x	1	1
0	1	0	x	x	0	x	x	x	x	x	0	0
0	1	0	x	x	1	x	x	x	x	x	1	1
0	1	1	x	x	x	0	x	x	x	x	0	0
0	1	1	x	x	x	1	x	x	x	x	1	1
1	0	0	x	x	x	x	0	x	x	x	0	0
1	0	0	x	x	x	x	1	x	x	x	1	1
1	0	1	x	x	x	x	x	0	x	x	0	0
1	0	1	x	x	x	x	x	1	x	x	1	1
1	1	0	x	x	x	x	x	x	0	x	0	0
1	1	0	x	x	x	x	x	x	1	x	1	1
1	1	1	x	x	x	x	x	x	x	0	0	0
1	1	1	x	x	x	x	x	x	x	1	1	1

CamScanner

Truth Table.

S_2	S_1	S_0	Y (Output)
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

Hence, The Multiplexer circuit was verified and output was observed.

Objective 2: Building a De-Multiplexer Circuit Using NOT, AND Gate.

Truth Table

Input	Select Input			Expected Output								Observed Output							
D_0	S_2	S_1	S_0	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1

CamScanner

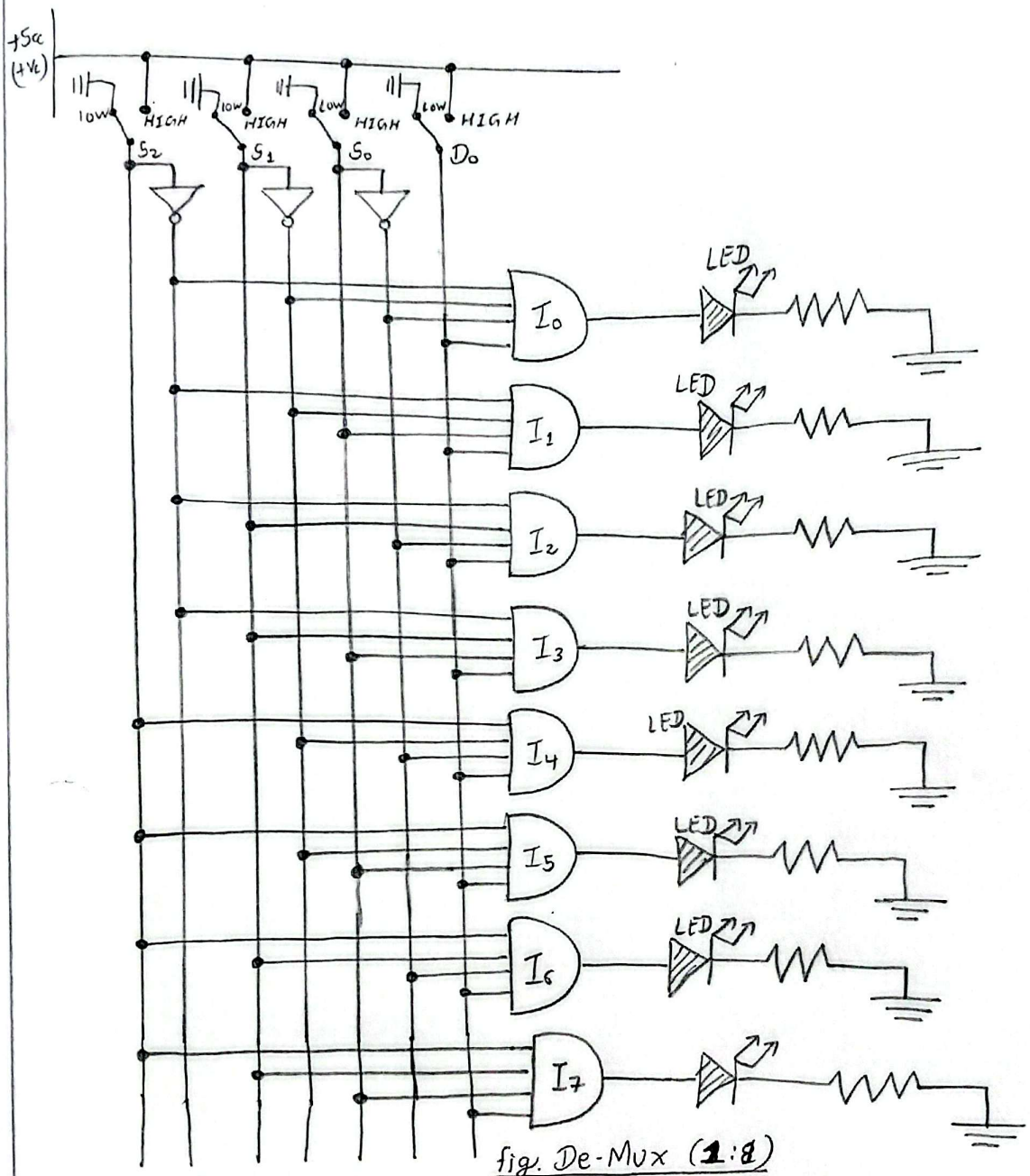


fig. De-Mux (1:8)

Hence, The De-Multiplexer circuit's expected and observed output are the same.

Discussion and Conclusion:

In this lab session, we verified the functionality of Multiplexer and De-Multiplexer circuits using different logic gates and truth tables.